AQA 3.2.4 Properties of Period 3 elements and their oxides

AQA A-level Chemistry · Revision questions · Sheet 4 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- P_4O_{10}, SO_2, SO_3 are simple molecular and acidic: $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 \rightarrow H_2SO_3$; $SO_3 \rightarrow H_2SO_4$
- Burning in O_2: Na yellow flame; Mg brilliant white light; P white flame and white smoke; S blue flame
- $Mg + H_2O(g) \rightarrow MgO + H_2$ (steam); $2Na + 2H_2O \rightarrow 2NaOH + H_2$
- Basic oxides react with acids to give salt + water; acidic oxides react with bases (e.g. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$)
- Na_2O and MgO are giant ionic and basic: $Na_2O + H_2O \rightarrow 2NaOH$ (pH 12–14); $MgO + H_2O \rightarrow Mg(OH)_2$ (pH 9–10)

Section A: Short-answer questions

01 Give an ionic equation for the reaction of oxide ions with water.

[1 mark]

02 Give an equation for the reaction of phosphorus(V) oxide with sodium hydroxide solution.

[1 mark]

03 Give an equation for the reaction of aluminium with oxygen.

[1 mark]

04 Which products are formed when magnesium reacts with steam?

[1 mark]

A Magnesium hydroxide and hydrogen
B Magnesium hydroxide and oxygen
C Magnesium oxide and hydrogen
D Magnesium oxide and oxygen

05 Sulfur burns in oxygen. Give an equation for this reaction and state one observation.

[2 marks]

06 Explain, in terms of structure and bonding, why sodium oxide forms an alkaline solution in water but phosphorus(V) oxide forms an acidic solution.

[2 marks]

Section B: Longer questions

07 This question is about oxides.

07.1 Sodium oxide forms a solution with a higher pH than magnesium oxide when equal amounts, in moles, of each oxide are added separately to equal volumes of water. State why both oxides form alkaline solutions. Suggest why sodium oxide forms a solution with a higher pH than the solution formed from magnesium oxide.

[2 marks]

07.2 Give an equation for the reaction between phosphorus(V) oxide and water.

[1 mark]

08 The elements sodium to sulfur in Period 3 all react with oxygen to form oxides.

08.1 Give an equation and two observations made for the reaction that occurs when sodium is heated in oxygen.

[2 marks]

08.2 Give an equation and one observation made for the reaction that occurs when phosphorus is heated in oxygen.

[2 marks]

09 This question is about some Period 3 elements and their oxides.

09.1 Write an equation for the reaction of phosphorus with an excess of oxygen.

[1 mark]

09.2 Describe a test you could carry out in a test tube to distinguish between sodium oxide and phosphorus(V) oxide.

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	$O^{2-} + H_2O \rightarrow 2OH^-$	*Source: Practice question (written for this set)*	1
02	$P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$	*Source: 7405/1 Jun 2022 Q05.5* Allow formation of acid salts, e.g. $P_4O_{10} + 8NaOH \rightarrow 4Na_2HPO_4 + 2H_2O$	1
03	$4Al + 3O_2 \rightarrow 2Al_2O_3$	*Source: Practice question (written for this set)*	1
04	**C**	*Source: 7405/3 Jun 2018 Q19*	1
05	$S + O_2 \rightarrow SO_2$ blue flame	*Source: Practice question (written for this set)*	2
06	M1 Na_2O is ionic and contains O^{2-} ions which react with water to form OH^- ions M2 P_4O_{10} is (simple) molecular / covalent and reacts with water to form an acid (H_3PO_4) that releases H^+ ions	*Source: Practice question (written for this set)*	2
07.1	M1 (oxide ions react with water to) form hydroxide ions / $O^{2-} + H_2O \rightarrow 2OH^-$ M2 sodium hydroxide is more soluble than magnesium hydroxide	*Source: 7405/3 Nov 2020 Q02.1–02.2* Ignore non-ionic equations for M1	2
07.2	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	Not P_2O_5	1
08.1	$4Na + O_2 \rightarrow 2Na_2O$ yellow / orange flame **and** white solid / powder / smoke	*Source: 7405/3 Jun 2018 Q02.1–02.2* Allow $2Na + O_2 \rightarrow Na_2O_2$ Do not allow ppt	2
08.2	$4P + 5O_2 \rightarrow P_4O_{10}$ white flame / light **or** white fumes / smoke / solid	Allow P_4O_6 Do not allow equations with P_2O_5	2
09.1	$P_4 + 5O_2 \rightarrow P_4O_{10}$	*Source: 7405/1 Jun 2017 Q06.1–06.2* Allow $4P + 5O_2 \rightarrow P_4O_{10}$; ignore state symbols	1
09.2	M1 react with water / add water M2 add litmus / universal indicator / measure pH M3 litmus: blue with sodium oxide (solution), red with phosphorus oxide (solution) / UI: blue-purple and red / sodium oxide pH 12–14, phosphorus oxide pH –1 to 2	If no M1 then CE = 0/3 M3 dependent on M2 Allow other reagents in solution that give a positive result, e.g. sodium carbonate solution	3